FINPRO — Project Synopsis

> **Cloud-Based Personal Finance & Investment Decision Support Platform**
> Final Year B.Tech · Computer Engineering · 2026–2027

Tagline: *Manage Today. Build Tomorrow.*
Keywords: PLAN · TRACK · ANALYZE · INVEST · GROW

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1. Introduction

Personal financial management is essential for individuals to achieve financial stability and long-term goals. **FinPro** is a cloud-based platform that integrates modern technologies such as AI, real-time market data, and secure cloud infrastructure to provide a comprehensive and personalized financial planning experience.

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2. Problem Statement

Most existing financial tools are fragmented, lack personalization, and do not provide holistic guidance across budgeting, investments, loans, and long-term planning. There is a need for an intelligent, unified, and scalable platform that can analyze personal financial data and provide actionable insights for better decision-making.

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3. Objectives

- Develop a full-stack personal finance management platform.
- Integrate AI for personalized financial insights and recommendations.
- Provide real-time market data for investment analysis.
- Implement secure and scalable cloud infrastructure.
- Enable tools for budgeting, goal setting, loan management, and financial planning.

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4. Scope

- Individual user financial management (personal use).

- Budgeting, expense tracking, and financial goals.
- Investment analysis (equity, mutual funds, gold, etc.).
- Loan & EMI tracking.
- AI-based financial advisor and what-if simulation.
- Cloud deployment with high availability and security.
- Web application with responsive UI.

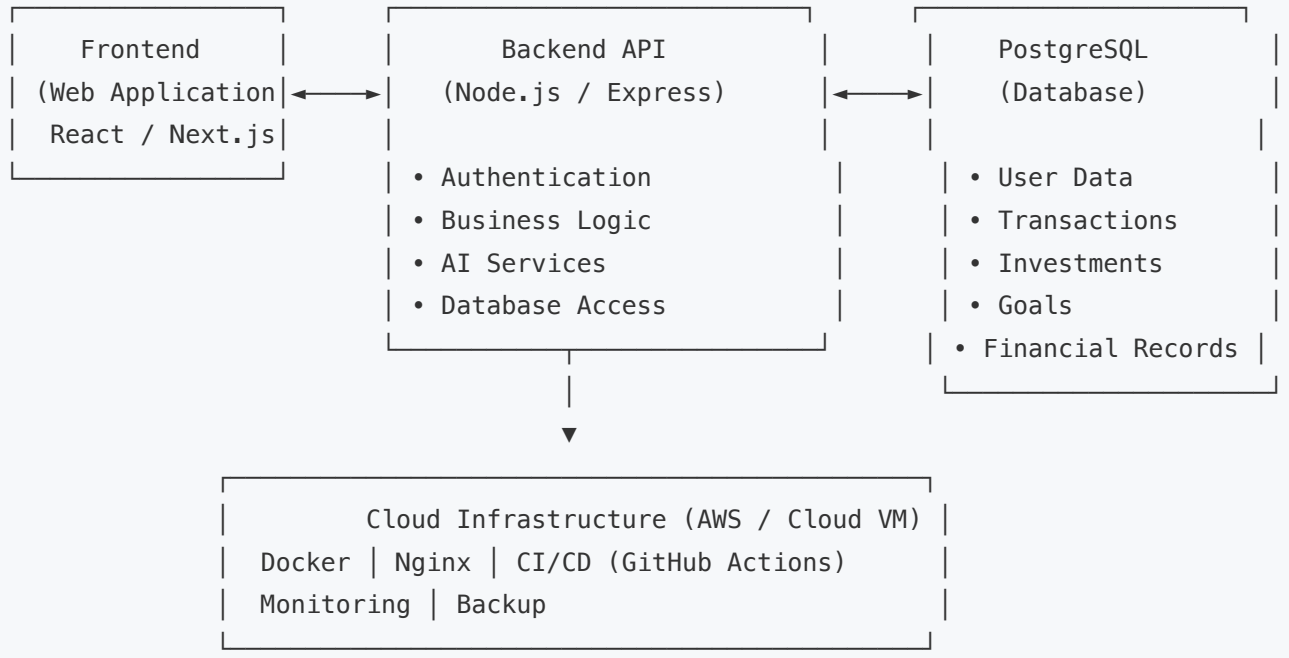
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5. Methodology

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Requirement Analysis
        ↓
System Design & Architecture
        ↓
Development (Frontend + Backend)
        ↓
Integration (AI, Market Data, Cloud)
        ↓
Testing & Deployment
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Evaluation & Documentation
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6. System Architecture



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7. Key Features

Feature	Description
📊 **Expense Tracking & Budgeting**	Log and categorize daily expenses with smart budgets
🎯 **Financial Goals & Planning**	Set, track, and achieve savings and investment goals
📈 **Investment Analysis & Portfolio Tracking**	Real-time equity, mutual fund, and gold portfolio
💳 **Loan & EMI Management**	Track active loans and EMI schedules
🤖 **AI Financial Advisor & What-If Analysis**	Gemini-powered advisory with scenario simulation
🧾 **Receipt OCR & Bank Statement Import**	Tesseract OCR for receipt scanning and parsing
🎙️ **Voice Assistant & Notifications**	Hands-free expense logging via Web Speech API
📰 **Real-Time Market Data & Financial News**	NSE/BSE equities, gold, FX rates & news feed
☁️ **Secure Cloud Deployment & Data Backup**	GCP Cloud Run with automated backups

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8. Expected Outcomes

- A fully functional and deployable **FinPro** platform.
- Improved financial awareness and decision-making for users.
- Demonstration of cloud computing, AI, and full-stack development skills.
- A scalable architecture for future enhancement.

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9. Technology Stack

Layer	Technology
Frontend	React / Next.js, Tailwind CSS
Backend	Node.js, Express.js
Database	PostgreSQL, Prisma ORM
AI / ML	OpenAI / LLM APIs, Google Gemini 1.5 Flash
Cloud	AWS / Docker / Nginx / GCP Cloud Run
CI/CD	GitHub Actions
Others	Real-time Market APIs, OCR, Voice, Notifications

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10. Conclusion

FinPro aims to provide an intelligent, secure, and scalable solution for personal financial management by leveraging cloud computing and AI. The project bridges the gap between traditional financial tools and modern technology, enabling users to make informed financial decisions and achieve their goals.

> *"Smarter Finances for a Brighter Tomorrow"*
